Synthesis Table — Android TPL Detection & SBOM Tools

Legend

Strong / Best-in-class ★	Strong	Partial / Limited	Weak / No	N/A / Not tested

criteria definitions

Obfuscation coverage — which specific techniques are handled: RN (renaming), PF (pkg flattening), CFR (control-flow randomization), DCR (dead-code removal), SE (string encryption), MI (method inlining / R8-specific).

Module decoupling — how the tool separates library code from app code. Package-name heuristics are fragile; CDG-based approaches are robust.

Partial lib handling — does the tool produce a similarity score for partially-included libraries (due to dead-code elimination), or only a binary match?

SBOM output — does the tool emit a structured SBOM artifact (CycloneDX, SPDX, custom JSON), or only print library names?

Open source — is the tool publicly available and reproducible for use as a baseline in experiments?

Comparison table

Tool	Year	Detection method	Lib-level F1	Version-level F1	R8 resilience	Speed (mean/app)	Obfusc. coverage	Module decoupling	Partial lib	SBOM output	Open source
GENERATION 1 — HASH & PACKAGE-BASED (2016–2018)											
LibRadar	2016	Android API freq. hashing + package clustering. Whitelist-based. No decompilation.	**~77%** *Prec. 97.9% / Recall ~50%*	**None** *Clustering; no version labels*	**Partial** *~86% recall under DashO*	**~5.5 s ★** *Fastest; single-dex only*	**RN, PF** *Fails on CFR, DCR, SE, MI*	**Pkg name** *Each pkg subtree = 1 candidate*	**No** *Binary match only*	**No** *Prints names to stdout*	**Yes** *GitHub (archived)*
LibScout	2016	3-layer Merkle tree. Fuzzy method descriptors. Similarity score 0–1.	**~83%** *Best Gen-1. Prec. 97.4%*	**Moderate** *Group-1 F1 64%; drops to 12.7% on adjacent versions*	**Partial** *F1 ~1% on R8-obf-shr-opt*	**82 s** *2nd fastest. Profile gen 2.8 s/lib*	**RN only ★** *100% resilient to RN; fails PF (−75%), CFR, MI*	**Pkg + CHA** *Class hierarchy; no CDG*	**Score 0–1** *Similarity score for partial libs*	**Partial** *Version list; no SBOM format*	**Yes** *GitHub (active)*
LibD	2017	Homogeny graph (call/inherit/inclusion). CFG opcode hashing per basic block.	**High** *1,954 libs vs LibRadar's 670. FP rate 25% vs 60.6%*	**None** *Clustering; no version labels*	**Strong** *Opcode hashing survives RN, PF by design*	**Medium** *1.4M apps tested; no per-app timing published*	**RN, PF** *CFR, MI untested*	**Homogeny graph** *Call/inherit/inclusion; best Gen-1*	**No** *Instance hash; no score*	**No** *Library names only*	**Partial** *Code available; reproducibility limited*
ORLIS	2018	Two-stage: sdhash app-filter → ssdeep CFG class	**~62–67 %** *Lowest Gen-1. Prec.*	**Implicit** *CFG fine-grained;*	**Partial** *Best Gen-1*	**~1,438 s ✗** *Slowest; crashes on 16*	**RN, PF, CFR** *CFG survives structural changes; SE, MI untested*	**CFG roots** *Structure-based root detection; no CDG*	**Jaccard score** *Jaccard similarity for partial libs*	**No** *Class-level mapping only*	**Partial** *Published; hard to install (Soot env)*

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		matching. Jaccard similarity.	*71%, Recall 63%*	*not reported as version metric*	*obfusc. resilience; not tested on R8*	*GB+ apps*					
GENERATION 2 — CDG-BASED & VERSION-AWARE (2021)											
ATVHunter	2021	CDG module decoupling → coarse CFG filter → fine opcode fuzzy rolling hash. Feature weighting.	**93.4% ★** *Prec. 98.6%, Recall 88.8%*	**88.8% ★** *Best at release. Prec. 90.6%, Recall 87.2%*	**Partial** *DashO tested; not explicitly on R8*	**66 s** *Two-stage filter; avoids full DB scan*	**RN, PF, CFR, DCR** *99.3% det. rate under RN+PF; −24% DCR; MI not tested*	**CDG ★** *Class Dependency Graph; best among reviewed tools*	**Fuzzy score** *Rolling hash gives % similarity for partial libs*	**Partial** *CVE-linked version list; no std SBOM format*	**Partial** *Online service; source partially released*
GENERATION 3 — R8-AWARE & SBOM-ORIENTED (2024–2025)											
LibMD	2025	Multi-scale fusion: class + function call + basic block dependency graphs. Bayesian version scoring.	**High** *Exceeds single-scale baselines; exact score unverified †*	**Strong** *Bayesian score targets adjacent-version disambiguation*	**Strong** *Dependency graph stable under R8 shrinking & inlining*	**N/A †** *Full paper unavailable*	**RN, PF, CFR, MI** *Multi-scale fusion designed for R8 inlining*	**Multi-scale CDG ★** *3-level dependency fusion; most robust approach*	**Prob. score** *Bayesian confidence for partial lib matches*	**Explicit SBOM ★** *Primary goal: SBOM generation + CVE mapping*	**No** *Not publicly released (2025)*
LibAttention	2025	Pre-train Transformer on 200M Smali tokens → fine-tune binary classifier. SDK structural vocabulary.	**~39%** *R8-non dataset; below top tools on non-obfusc.*	**None** *Binary presence/absence only; no version label*	**Best overall ★** *+30% F1 over all rule-based under R8-obf-shr-opt*	**~472 s** *AndroGuard preprocessing dominates*	**All R8 modes ★** *Deep learning learns obfusc. patterns: RN, PF, CFR, DCR, SE, MI*	**Implicit** *Model learns boundaries; not explicit CDG*	**No** *Binary output; no score for partial libs*	**No** *No version → no CVE → no SBOM*	**Partial** *Code described; full release unclear*
SAD	2025	CDG neighbor aggregation → stateful/stateless class split → opcode + field-op-seq matching → literal tie-breaker.	**97.6% ★** *Highest lib-level F1 of all reviewed tools*	**84.8% ★** *+10.98 % over LIBLOOM. Class-level F1 94.1%*	**Strong** *CDG resilient to R8 renaming & shrinking; −10pp under heavy inlining*	**~130 s** *Slower than LIBLOOM (5 s), LibScan (46 s)*	**RN, PF, CFR, DCR** *Weaker on MI (method inlining, R8-opt)*	**CDG + stateful/stateless ★** *Most granular module decoupling reviewed*	**Class score** *Per-class confidence; handles partial inclusion*	**Partial** *Version labels produced; needs NVD integration*	**Partial** *EASE 2025; availability unclear at submission*